Previewing the USB Camera Feed

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1. Implementation Principle

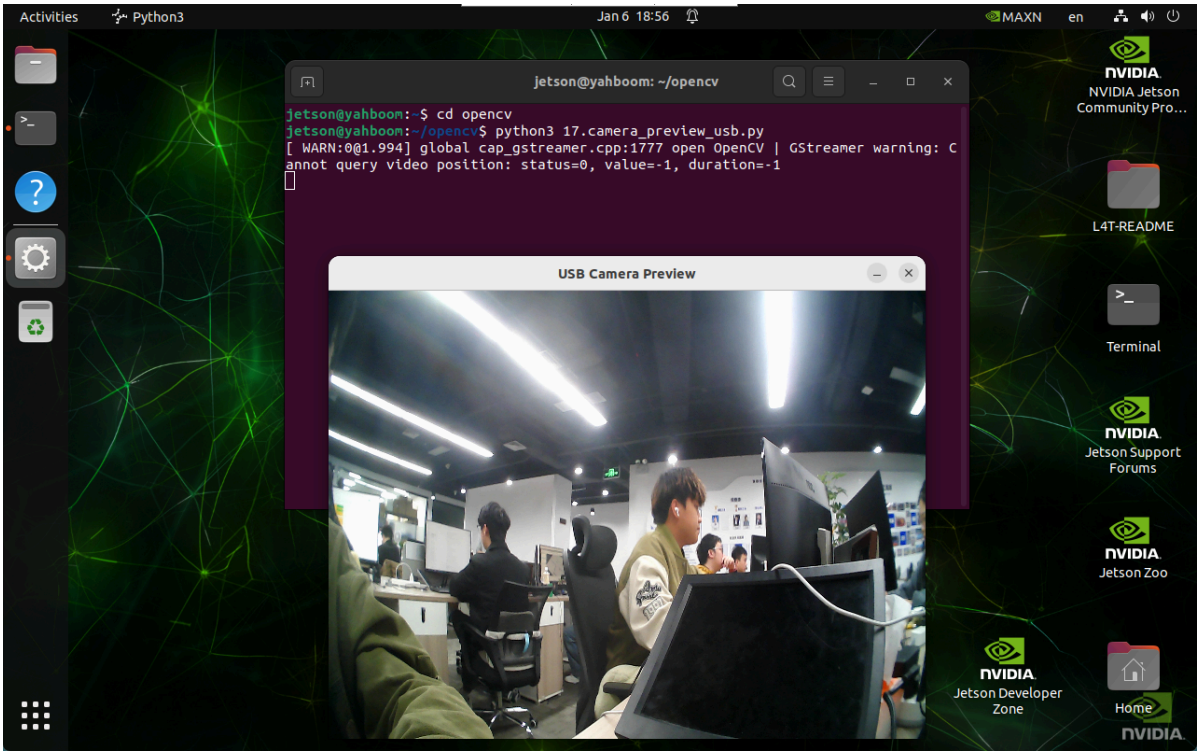
Use the `cv2.VideoCapture()` function to capture video frames from a USB camera or a video file.

2. Implementation Effect

```
cd ~/opencv
```

```
python3 17.camera_preview_usb.py # The program opens video0 by default; to open another camera, modify the device number yourself
```

Note: click the camera window and press `q` to exit.



3. Implementation Code

```
import cv2

def preview_usb_camera():
    cap = cv2.VideoCapture(0)
    cap.set(6, cv2.VideoWriter.fourcc('M', 'J', 'P', 'G'))
    cap.set(cv2.CAP_PROP_FRAME_WIDTH, 640)
    cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)
    if not cap.isOpened():
```

```
        print("Error: Could not open USB camera.")
        return

    while True:
        ret, frame = cap.read()
        if not ret:
            break

        cv2.imshow('USB Camera Preview', cv2.resize(frame, (640, 480)))
        if cv2.waitKey(1) & 0xFF == ord('q'):
            break

    cap.release()
    cv2.destroyAllWindows()

preview_usb_camera()
```